

Eluri Karthik

Senior Full-Stack Python Developer | Cloud-Native Microservices
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PROFESSIONAL SUMMARY

- Accomplished Senior Python Full-Stack Developer with 9+ years of experience designing, developing, and deploying enterprise-grade web applications and microservices across diverse domains. Demonstrated expertise in full-stack development, cloud-native architecture, event-driven systems, and CI/CD automation, with deep hands-on experience in Python frameworks, modern frontend technologies, distributed databases, and observability tooling. Proven ability to lead technical design, optimize high-throughput systems, and mentor development teams.
- Advanced experience in Python 3.x, designing and implementing Django (1.x–4.x), FastAPI, and Flask microservices to build secure, high-performance, and scalable backend systems with maintainable code architecture.
- Skilled in RESTful APIs, GraphQL, gRPC, WebSockets, and implementing JWT/OAuth2 authentication within event-driven microservices, enabling low-latency, high-throughput inter-service communication and robust distributed architectures.
- Proficient in React.js (16–18+), Next.js, Redux, TypeScript, SCSS, Tailwind CSS, Bootstrap, and Webpack, creating interactive, responsive, and enterprise-grade dashboards with real-time data visualization, state management, and performance optimization.
- Deep knowledge of relational, NoSQL databases, including PostgreSQL (10–14), MySQL, MongoDB, Cassandra, ElasticSearch, Azure Cosmos DB, with optimized schema design, indexing, and query performance.
- Hands-on experience with cloud-native platforms including GCP (GKE Autopilot, Cloud Build, Pub/Sub), Azure (AKS, Functions, DevOps, App Insights), and AWS (ECS, RDS, S3) for deploying, monitoring, scaling full-stack microservice applications.
- Advanced containerization and orchestration expertise using Docker, Kubernetes, Helm, ArgoCD, and Terraform, leveraging GitOps workflows for reproducible, fault-tolerant, and scalable deployments.
- Applied asynchronous and event-driven processing using Celery, Redis Queues, and Google Pub/Sub to orchestrate high-throughput background tasks, distributed workflows, and near real-time event handling across microservices.
- Enforced TDD/BDD methodologies leveraging pytest, unittest, Selenium, Postman, Newman, Coverage.py, and Pact, ensuring maintainable, contract-compliant, and production-grade microservices with automated regression safety.
- Implemented full-stack observability and performance monitoring with Prometheus, Grafana, ELK Stack, Sentry, and distributed tracing, enabling proactive SLA adherence, real-time alerts, and end-to-end workflow visibility.
- Established cloud security and secrets management using Google Secret Manager, Azure Key Vault, and IAM-based access controls, enforcing enterprise compliance and protecting sensitive distributed workloads.
- Engineered optimized PostgreSQL and Cassandra schemas with Redis-backed caching, advanced indexing, and query pipelines, delivering low-latency access, high-throughput analytics, and scalable real-time systems.
- Refactored legacy systems to modern microservice architectures, including FastAPI migration, async endpoints, dependency injection, and event-driven patterns, improving maintainability, throughput, and scalability.
- Developed serverless and event-driven microservices using Azure Functions and Pub/Sub triggers, enabling near real-time processing, asynchronous workflows, and seamless cross-service communication.
- Proficient in collaborative development and release management using Git, GitHub, GitLab, Jira, and shell scripting, ensuring version control integrity, traceability, and streamlined delivery pipelines.

TECHNICAL SKILLS

- Programming Languages & Frameworks:** Python 3.x, Django (1.x-4.x), FastAPI, Flask, JavaScript ES6, TypeScript, React.js (16-18+), Next.js, Redux
- Databases & Caching:** PostgreSQL (10-14), MySQL, MongoDB, Redis, Cassandra, ElasticSearch, Azure Cosmos DB
- Cloud & Infrastructure:** GCP (GKE Autopilot, Cloud Build, Secret Manager, Pub/Sub), AWS (EC2, RDS, S3, IAM), **Azure** (AKS, Functions, ARM Templates, Cosmos DB, DevOps, App Insights, Containers)
- DevOps & Containerization:** Docker, Kubernetes, Helm, ArgoCD, Terraform, GitOps, Jenkins, Nginx, Gunicorn
- Microservices & APIs:** REST API, GraphQL, gRPC, WebSockets, JWT, OAuth2, Event-driven Architecture, Async Endpoints
- Frontend Technologies:** HTML5, CSS3, SCSS, Tailwind CSS, Bootstrap (3-4), Webpack
- Testing & Quality Assurance:** pytest, unittest, Selenium, Postman, Newman, Coverage.py, Pact, TDD/BDD
- Monitoring & Observability:** Prometheus, Grafana, ELK Stack, Sentry, Distributed Tracing
- Message Queues & Background Processing:** Celery, Redis Queues, Google Pub/Sub
- Tools & Version Control:** Git, GitHub, GitLab, Jira, Shell Scripting

WORK EXPERIENCE

Senior Full-Stack Python Developer | Cardinal Health – Dublin, Ohio, US

Jan 2025 – Present

Project: Intelligent Healthcare Workflow Orchestration Platform (IHOP)

Overview: Served for the end-to-end design and development of a cloud-native, event-driven healthcare workflow orchestration platform for Cardinal Health, enabling real-time coordination of pharmacy orders, claims processing, and clinical inventory management across distributed systems. The platform unified transactional and analytical data streams through Django 4.x, FastAPI microservices, and gRPC-based communication, ensuring low-latency, fault-tolerant data exchange between internal applications and external partners.

- Developed Django 4.x and FastAPI microservices leveraging PostgreSQL, Redis, and Cassandra to orchestrate high-throughput, low-latency healthcare workflows, ensuring data consistency, scalability, and reliability across distributed systems.

- Implemented gRPC-based inter-service communication and event-driven microservices, enabling scalable, low-latency message exchange, real-time workflow orchestration, and cross-service data consistency.
- Developed Next.js and React 18+ frontends with Tailwind CSS, integrating WebSocket-based real-time updates for dashboards, live notifications, and interactive healthcare analytics.
- Architected event-driven pipelines using Google Pub/Sub, supporting streaming ingestion, asynchronous processing, and near real-time notification delivery for high-throughput healthcare events.
- Optimized Cassandra and PostgreSQL schemas, partitioning, indexing, and query strategies, ensuring efficient handling of high-volume transactions and complex analytical workloads.
- Containerized Python microservices with Docker and orchestrated production deployments using Kubernetes and Helm charts, enabling reproducible, fault-tolerant, and scalable cloud-native environments.
- Automated CI/CD pipelines leveraging ArgoCD, Terraform, and GitOps workflows on GKE Autopilot, supporting automated testing, environment provisioning, and seamless production deployments.
- Enforced TDD/BDD practices using unittest, Coverage.py, and Pact, ensuring high-quality code, automated contract validation, and maintainable, production-ready microservices.
- Configured Prometheus and Grafana dashboards to monitor microservice performance, API latency, system resource utilization, and integrated alerting pipelines for proactive incident detection.
- Integrated ELK Stack and Sentry for centralized logging, distributed tracing, and real-time error monitoring, enabling rapid root-cause analysis and reducing mean-time-to-resolution.
- Managed secrets and sensitive configurations using Google Secret Manager, implementing role-based access controls and enforcing enterprise compliance standards across all environments.
- Refactored legacy Python services to FastAPI with async endpoints, enhancing service scalability, maintainability, and request throughput for high-concurrency workloads.
- Collaborated with cross-functional teams to design RESTful APIs, enforce API versioning, and establish data contracts, ensuring consistent integration and interoperability across microservices.
- Developed asynchronous background jobs and caching strategies with Redis, optimizing high-throughput workflows, reducing response times, and improving overall system efficiency.
- Applied advanced observability and monitoring best practices, integrating Prometheus, Grafana, ELK metrics, and distributed tracing into alerting pipelines for proactive system health management.
- Leveraged Google Cloud Build and GKE Autopilot to streamline CI/CD pipelines, implementing automated testing, security checks, and zero-downtime deployments.

Environment: Python 3.x, Django 4.x, FastAPI, gRPC, Event-driven microservices, React 18+, Next.js, Tailwind CSS, PostgreSQL 14, Redis, Cassandra, Docker, Kubernetes, Helm, ArgoCD, Terraform, GitOps, Prometheus, Grafana, ELK Stack, Sentry, TDD/BDD, Pact, Coverage.py, Google GKE Autopilot, Google Cloud Build, Google Secret Manager, Google Pub/Sub, REST API, Postman, unittest, JWT, OAuth2, WebSockets, API versioning, distributed tracing, async endpoints

Senior Full-Stack Python Developer | The Cigna Group – Houston, TX

Aug 2023 – Dec 2024

Project: Claims Intelligence & Care Coordination Platform

Overview: Contributed for the development of a cloud-native, full-stack healthcare platform to modernize Cigna's claims lifecycle management, patient-care coordination ecosystem. The platform unified claim intake, eligibility verification, adjudication, real-time analytics into a microservices-driven architecture, ensuring seamless interoperability across multiple data systems, full compliance with HIPAA and FHIR standards.

- Designed and implemented GraphQL endpoints to aggregate and resolve multi-source datasets, enabling flexible queries, efficient data retrieval, and seamless integration with frontend analytics dashboards.
- Developed FastAPI and Django 3.x microservices leveraging PostgreSQL and Redis to orchestrate high-throughput health claim processing, ensuring real-time data consistency, robust transaction handling across distributed healthcare systems.
- Built React 17+ frontends with Redux and TypeScript, incorporating WebSockets and dynamic state management to deliver real-time dashboards, notifications, and interactive user interfaces.
- Implemented asynchronous pipelines with Celery and Redis queues, orchestrating event-driven background tasks for data enrichment, notifications, and high-throughput processing workflows.
- Optimized PostgreSQL indexing strategies and Elasticsearch queries, improving latency and throughput for complex analytics, search functionality, and large-scale data operations.
- Containerized applications using Docker and deployed via Kubernetes on Azure AKS, leveraging Helm charts to ensure reproducible, scalable, and production-ready environments.
- Automated CI/CD pipelines with GitHub Actions and Terraform, including infrastructure provisioning, environment configuration, and continuous delivery for cloud-native deployments on AKS.
- Developed comprehensive unit and integration tests with pytest and validated APIs using Postman and Newman, ensuring system reliability, regression safety, and strict adherence to API contracts.
- Built serverless Azure Functions triggered by events and pub/sub messaging, enabling real-time updates, asynchronous workflows, and seamless integration across multiple microservices.
- Managed secure, distributed, multi-region data storage using Azure Cosmos DB, ensuring transactional consistency, low-latency access, and high availability for critical healthcare workloads.
- Configured Prometheus and Grafana dashboards to monitor application performance, API latency, system metrics, and SLA compliance, enabling proactive incident detection and resolution.

- Refactored legacy Django endpoints to FastAPI with asynchronous support, enhancing request throughput, service maintainability, and scalability for high-concurrency workflows.
- Applied frontend optimization strategies with SCSS, TypeScript, and React performance patterns, improving rendering speed, maintainability, and user experience for enterprise dashboards.

Environment: Python 3.8+, FastAPI, Django 3.x, Celery, REST APIs, GraphQL, React 17+, Redux, TypeScript, SCSS, PostgreSQL 12, Redis, Elasticsearch, Docker, Kubernetes, GitHub Actions, Terraform, Prometheus, Grafana, pytest, Postman, Newman, Azure AKS, Azure Functions, Azure ARM Templates, Azure Cosmos DB

Python Full-Stack Developer | McKinsey & Company – New York City, NY

Mar 2021 – Jul 2023

- Developed Django 2.x and FastAPI microservices with RESTful APIs, JWT authentication, and OAuth2 flows, powering internal analytics and reporting platforms with secure access and standardized service contracts.
- Built React.js 16+ dashboards with Redux and Webpack, integrating WebSockets and dynamic data binding to provide real-time, interactive, and enterprise-grade user interfaces.
- Implemented asynchronous task pipelines using Celery, integrating Redis queues and PostgreSQL, orchestrating high-volume data processing and event-driven workflows with reliability and fault tolerance.
- Designed and optimized PostgreSQL schemas, indexing strategies, and Redis caching layers, reducing query latency and supporting high-concurrency workloads for internal analytics.
- Managed MongoDB collections and aggregation pipelines for semi-structured and analytical datasets, enabling flexible reporting, fast lookups, and complex data transformations.
- Containerized and orchestrated applications with Docker and Azure Containers, implementing environment-specific configurations, automated deployments, and reproducible cloud-native workflows.
- Leveraged Azure DevOps, App Insights, and custom logging frameworks for observability, tracking API metrics, detecting anomalies, and alerting critical workflow deviations in real-time.
- Refactored legacy Flask microservices to FastAPI, introducing async endpoints, dependency injection, and request validation, enhancing maintainability, request throughput, and developer productivity.
- Coordinated cross-team API design with REST standards, versioning, and OpenAPI specifications, enforcing consistent contracts, backward compatibility, and integration compliance.
- Implemented advanced logging, distributed tracing, and performance monitoring using Azure App Insights, Redis, and custom middleware, enabling proactive system observability and end-to-end workflow monitoring.

Environment: Python 3.6+, Django 2.x, FastAPI, Flask, Celery, React.js 16+, Redux, Webpack, PostgreSQL 10+, Redis, MongoDB, Docker, Jenkins, GitLab CI/CD, Nginx, pytest, Selenium, Azure DevOps, Azure Containers, Azure App Insights

Python Web Developer | Fractal Analytics – Mumbai, India

Jul 2018 – Oct 2020

- Developed Django 2.x web applications with REST APIs to power analytics dashboards and internal reporting tools.
- Designed and optimized PostgreSQL schemas and queries, improving data retrieval efficiency for large datasets.
- Implemented backend microservices in Flask for lightweight data processing and asynchronous tasks.
- Built dynamic frontend components using JavaScript ES6 and Bootstrap 4, delivering responsive and interactive user interfaces.
- Developed and maintained MongoDB collections for semi-structured datasets used in analytics workflows.
- Deployed applications on AWS EC2 instances with Nginx/Gunicorn, ensuring secure and scalable production environments.
- Managed AWS RDS and S3 for structured data storage and ETL pipeline outputs, ensuring reliability and persistence.
- Implemented unit and integration tests with pytest, validating backend functionality and API reliability.
- Created Postman collections for standardized API testing and cross-team collaboration.
- Leveraged Git, GitHub for version control, tracked tasks using Jira, ensuring smooth project management and code traceability.
- Automated deployment scripts and maintenance tasks using shell scripting, improving operational efficiency.
- Refactored legacy Django 1.11 applications to Django 2.x and modernized REST endpoints, improving maintainability and aligning with current Python standards.

Environment: Python 3.x, Django 2.x, Django REST Framework, Flask, JavaScript ES6, Bootstrap 4, PostgreSQL, MongoDB, Nginx, Gunicorn, Git/GitHub, pytest, Postman, AWS EC2/RDS/S3/IAM

Python Developer | EXL Service – Bengaluru, India

Mar 2016 – Jun 2018

- Developed Django 1.x and Flask web applications, implementing core business logic for internal enterprise workflows.
- Built and maintained REST APIs using Python frameworks, enabling seamless integration with MySQL databases.
- Designed and optimized MySQL schemas and queries to support structured data storage and reporting requirements.
- Implemented frontend features with HTML5, CSS3, JavaScript, Bootstrap 3, ensuring responsive UI components for internal tools.
- Refactored legacy Python 2.7 codebases to Python 3.x, improving maintainability and aligning with modern Python standards.
- Version-controlled all projects using Git, ensuring code traceability and collaboration across the development team.
- Developed unit tests with unit tests, validating backend logic and API responses to reduce production defects.
- Created Postman collections for API testing and documentation, standardizing backend service verification.
- Assisted in application deployment and debugging, coordinating with senior developers to troubleshoot runtime issues.
- Maintained technical documentation, coding standards, ensuring knowledge sharing, and smooth onboarding of new team members.

Environment: Python 3.x, Django 1.x, Flask, REST API, MySQL, HTML5, CSS3, JavaScript, Bootstrap 3, Git, unittest